Efficacy And Safety of Triple Fixed Dose Combination Of Dapagliflozin, Sitagliptin and Extended Release Metformin Tablets in Indian Patients With Type 2 Diabetes Mellitus Poorly Controlled With Metformin: Phase 3 Randomized, Open-label, Three-arm Study in Comparison With Dual Combinations

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T2DM – Scope of Problem and Unmet Medical Need

- **Dual therapy (DPP4i + Met/SGLT2i + Met)** with mean baseline HbA1c of 8.9% have shown **only 18%-22% are able to achieve target HbA1c < 7%**[1,2]
- **Weight gain and hypoglycaemia** – adversely affect **patient adherence & quality of life** – impacting glycaemic goals[3]
- **High pill burden** and complex treatment regimens → ↓ **adherence**[4] **– each 10% increase in OAD medication adherence was associated with a 0.1% HbA1c reduction ($p = 0.0004$)**
- **FDCs → improve patient compliance,** glycaemic control[5] and have potential to decrease risk of complications[6]
- Progressive loss of β-cell function, necessitate patients requiring **multiple OADs with differing MOA to achieve target HbA_{1c} levels**[7]

FDCs of drugs with complementary MOAs and β-cell preserving action – can improve compliance, provide effective glycaemic control, lead to weight loss without increased risk of hypoglycaemia [6,7]

1. Unnikrishnan R, et al. Nature Reviews Endocrinology 2016; 2. Julio Rosenstock et al. Dia Care 2015;38:376-383 3. Khunti K, *et al. Diabetes Obes Metab* 2010;**12**:474–84 4. Black JA, et al. BMJ Open Diabetes Research and Care 2015;3:e000075 5. Yurgin NR, et al. Patient Prefer Adherence 2008 6. Benford M, et al. Adv Ther 2012 7.Jabbour SA, et al. Diabètes Care 2014;37:740–750

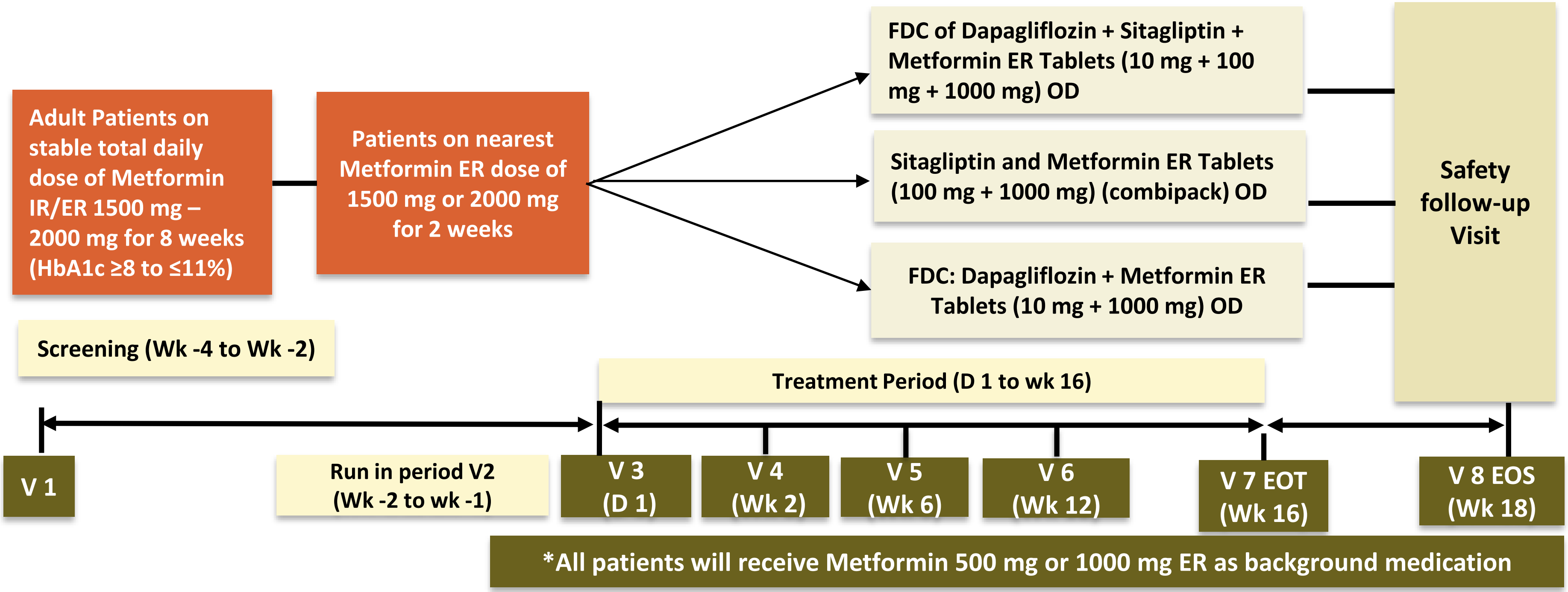
Fixed dose combination of Dapagliflozin + Sitagliptin + Metformin – Rationale

Parameter	Metformin	Dapagliflozin	Sitagliptin	Comments
Efficacy (HbA1c reduction)	0.9 - 1.3%	0.8 - 0.9%	0.7-0.8 %	**FDC can show improved efficacy & synergistic action**
Mechanism of action	Reduction of hepatic glucose output	Glucose excretion through urine	DPP4 inhibition to stimulate insulin release	**Complementary mechanisms of action of 3 drugs**
Dosing	1000 mg ER OD	10 mg OD	100 mg OD	**PK parameters matching and conducive to OD dosing**
Weight reduction	Weight loss	Weight loss	Weight neutral	**Weight loss/weight neutral**
Hypoglycemia	Minimal risk	Minimal risk	Minimal risk	**Minimal Risk**
β – cell Preservation	Insulin sensitizing action	Unique insulin independent mechanism	Glucose dependent incretin effect	**Combination has shown improvement in β – cell function**
CV outcome data	Proven cardiovascular safety	DECLARE-TIMI trial - Dapagliflozin 10 mg has shown additional cardiac and renal benefits	Proven cardiovascular safety shown in large TECOS outcomes study	**CV & renal benefits offered by Dapagliflozin will make the FDC beneficial to diabetics with comorbidities**

Thus Dapagliflozin plus Sitagliptin (10 + 100mg) + Metformin 1000 mg ER in a FDC would be useful for patients to achieve sustained glycaemic control with increased patient adherence by reducing their pill burden, with additional benefit of weight loss without increasing risk of hypoglycaemia

Objective and Study Design

Efficacy, safety and tolerability of **triple FDC of Dapagliflozin, Sitagliptin and Metformin** HCl ER tablets (10 mg + 100 mg + 1000 mg) OD in comparison to Co-administration of **Sitagliptin Phosphate 100 mg and Metformin HCl SR** tablets 1000 mg OD, and dual FDC of **Dapagliflozin 10 mg and Metformin HCl ER** tablets 1000 mg OD



Study Endpoints

Primary Endpoint:

- Mean change in HbA1c (Glycosylated hemoglobin) from baseline at the end of week 16

Secondary Endpoint:

Efficacy:

- Mean change in HbA1c (Glycosylated haemoglobin) from baseline at the end of week 12
- Mean change in Postprandial Blood Glucose (PPBG) from baseline at the end of week 12 and week 16
- Mean change in Fasting Blood Glucose (FBG) from baseline at the end of week 12 and week 16
- Proportion of Participants Achieving HbA1c < 7.0% at the end of week 12 and week 16
- Mean change in bodyweight from baseline to end of week 16
- Number of patients requiring rescue medication

Safety:

- Safety assessment includes Treatment Emergent Adverse Events (TEAEs) reported during the study
- Number of patients requiring hypoglycaemia management

Study Hypothesis and Sample Size Estimation

Study Hypothesis:

Superiority in efficacy and comparable safety of Triple FDC over dual combinations.

Sample Size Estimation

- With a power of 85%, alpha of 5%, difference in mean of 0.4% with a standard deviation of 1 was considered for sample size estimation
- Assuming 15% attrition, 134 patients in each arm were needed to be enrolled
- Total sample of 402 patients was found to be satisfactory to meet the study objectives

Statistical Analysis

- A MMRM model was used to test significant difference between test drug and comparator drugs
- The model had change in HbA1c, PPBG, FBG and body weight as the dependent variable and treatment, visit, treatment by visit interaction as fixed effects and baseline HbA1c as a covariate
- Proportion of participants achieving HbA1c < 7.0% at Week 12 and Week 16 and number of patients requiring rescue medication was summarized using counts and percentages
- Chi-square test (if expected frequency >5) was used to test significant difference between the two groups proportions
- Numbers of TEAEs and incidence rates were tabulated using available version of MedDRA version 24.0 by Preferred term (PT), System Organ Class (SOC), severity and relationship

Patient Disposition

	FDC of Dapagliflozin 10 mg, Sitagliptin 100 mg and Metformin HCl ER 1000 mg tablets (N = 137)	Sita 100 mg + Met SR 1000 mg (combipack tablet) (N=139)	FDC of Dapagliflozin 10 mg and Metformin HCl ER tablets 1000 mg (N = 139)	Overall (N = 415)
ITT Population, n (%)	137 (100%)	139 (100%)	139 (100%)	415 (100%)
mITT Population, n (%)	137 (100%)	139 (100%)	139 (100%)	415 (100%)
PP Population, n (%)	135 (98.5%)	133 (95.7%)	136 (97.8%)	404 (97.3%)
Safety Population, n (%)	137 (100%)	139 (100%)	139 (100%)	415 (100%)
Study Completion Status, n (%)				
Yes	135 (98.5%)	133 (95.7%)	136 (97.8%)	404 (97.3%)
No	2 (1.5%)	6 (4.3%)	3 (2.2%)	11 (2.7%)
Reason for discontinuation, n (%)				
Consent Withdrawn	0	1 (0.7%)	0	1 (0.2%)
Lost to follow up	2 (1.5%)	5 (3.6%)	3 (2.2%)	10 (2.4%)

ER = extended release; FDC = fixed dose combination; HCl = hydrochloride; ITT = intention to treat; mITT = modified intention to treat; PP = per protocol, SR = sustained release.

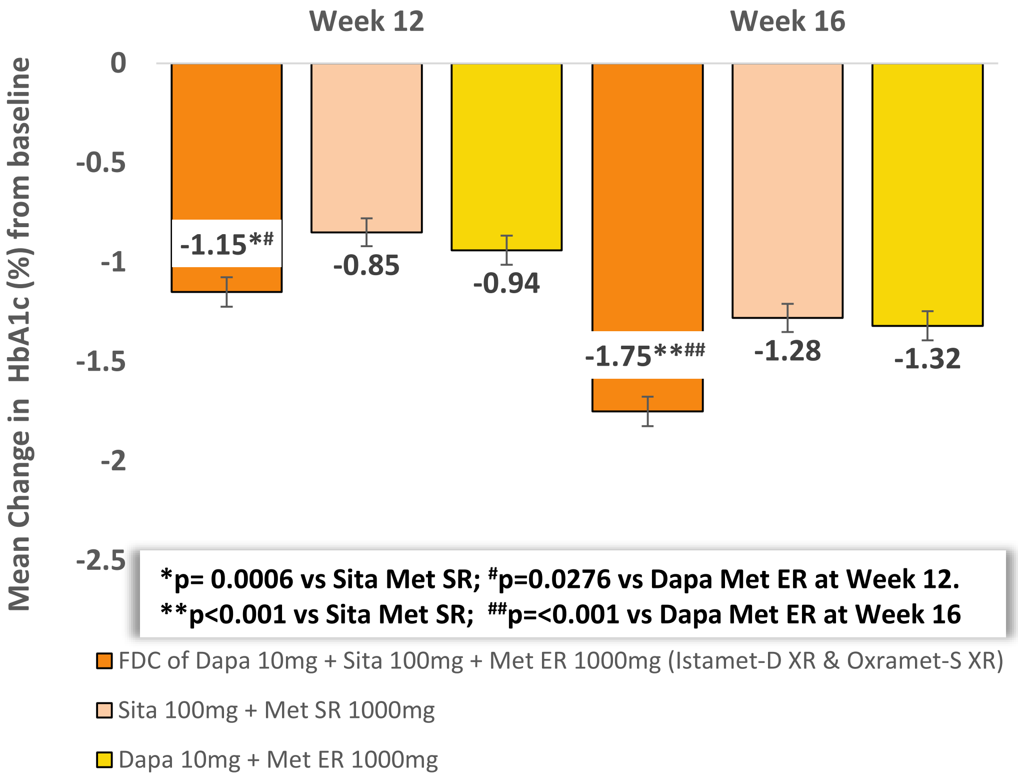
Baseline Characteristics

Parameters	FDC of Dapagliflozin 10 mg, Sitagliptin 100 mg and Metformin HCl ER 1000 mg tablets (N = 137)	Sita 100 mg + Met SR 1000 mg (combipack tablet) (N=139)	FDC of Dapagliflozin 10 mg and Metformin HCl ER tablets 1000 mg (N = 139)	Test Drug Vs Comparator 1 p-value	Test Drug Vs Comparator 2 p-value
Gender n(%)					
Male	82 (59.9%)	80 (57.6%)	87 (62.6%)	0.6980	**0.6409**
Female	55 (40.1%)	59 (42.4%)	52 (37.4%)		
Age (Years)					
Mean ± SD	49.09 ± 9.59	48.63 ± 10.53	49.13 ± 9.59	0.7081	**0.9711**
Height (cms)					
Mean ± SD	159.9 ± 8.81	160.6 ± 7.73	160.9 ± 8.88	0.5103	**0.3396**
Weight (kg)					
Mean ± SD	66.47 ± 10.98	67.88 ± 10.97	68.77 ± 11.49	0.2866	**0.0910**
Body Mass Index (kg/m^2)					
Mean ± SD	26.01 ± 3.97	26.35 ± 4.07	26.60 ± 4.45	0.4803	**0.2458**

Baseline characteristics are comparable in all treatment groups

Primary Endpoint: Mean Reduction in HbA1c (%) at Week 12 and Week 16

Mean reduction in HbA1c From Baseline



		FDC of Dapa 10 mg +Sita 100 mg +Met ER 1000 mg (N = 137)		Sita 100 mg + Met SR 1000 mg (combipack tablet) (N=139)		FDC of Dapa 10 mg + Met ER 1000 mg (N=139)	
Visits	**Statistic Summary**	**Actual**	**CFB**	**Actual**	**CFB**	**Actual**	**CFB**
Baseline	**n**	137		139		139	
	Mean ± SD	9.05 ± 0.87		8.99 ± 0.83		8.93 ± 0.86	
Week 12 (Visit 4)	**n**	135		133		137	
	Mean ± SD	7.90 ± 0.91	**-1.15 ± 0.87**	8.12 ± 0.92	**-0.85 ± 0.70**	8.00 ± 0.91	**-0.94 ± 0.73**
Week 16 (Visit 5)	**n**	130		130		129	
	Mean ± SD	7.30 ± 0.90	**-1.75 ± 0.92***	7.69 ± 0.91	**-1.28 ± 0.69**	7.62 ± 0.92	**-1.32 ± 0.74**
	p value	<0.0001		<0.0001		<0.0001	

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Secondary Endpoint: Mean Reduction in PPBG (mg/dL) From Baseline to Week 12 and Week 16

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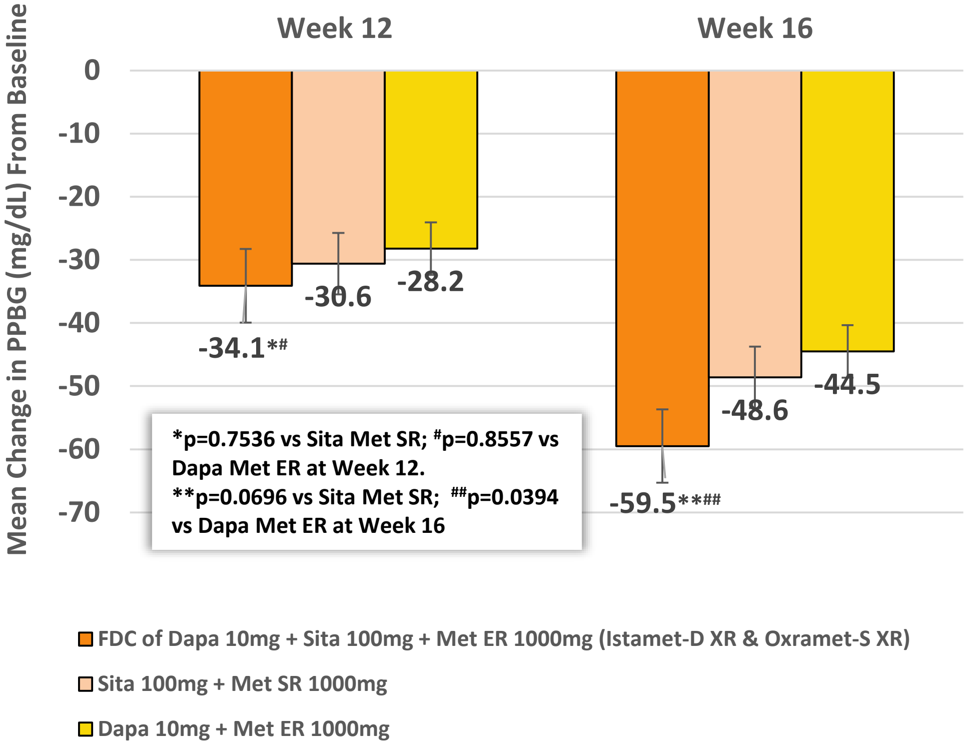
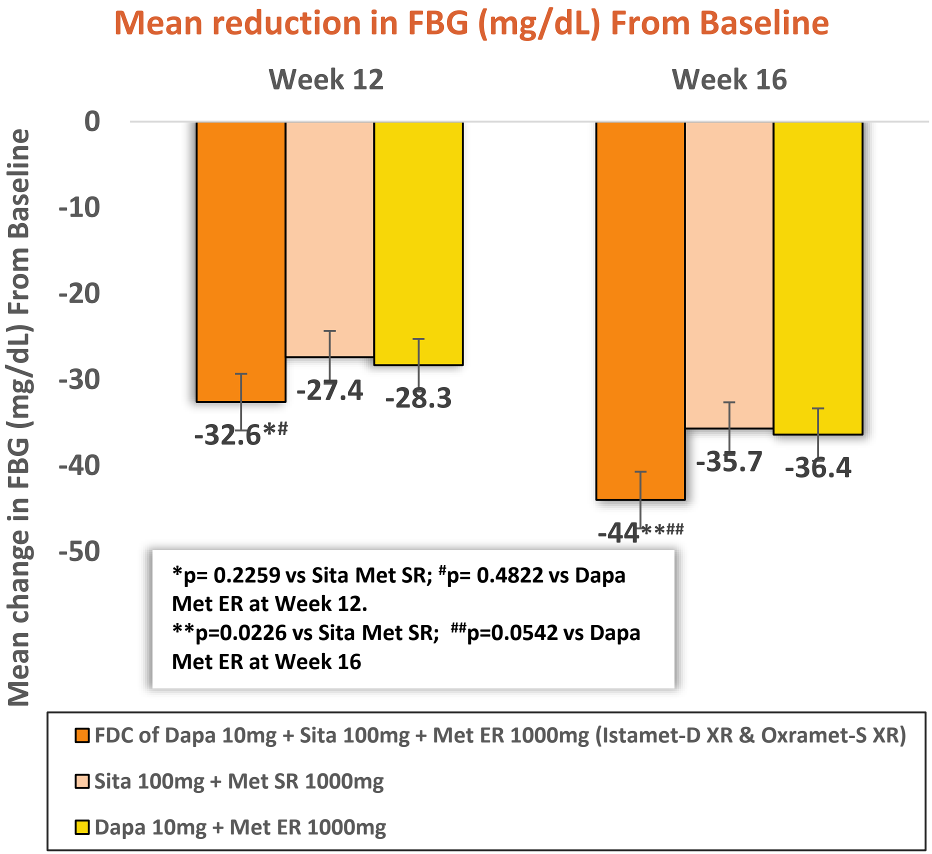
Mean reduction in PPBG (mg/dL) From Baseline



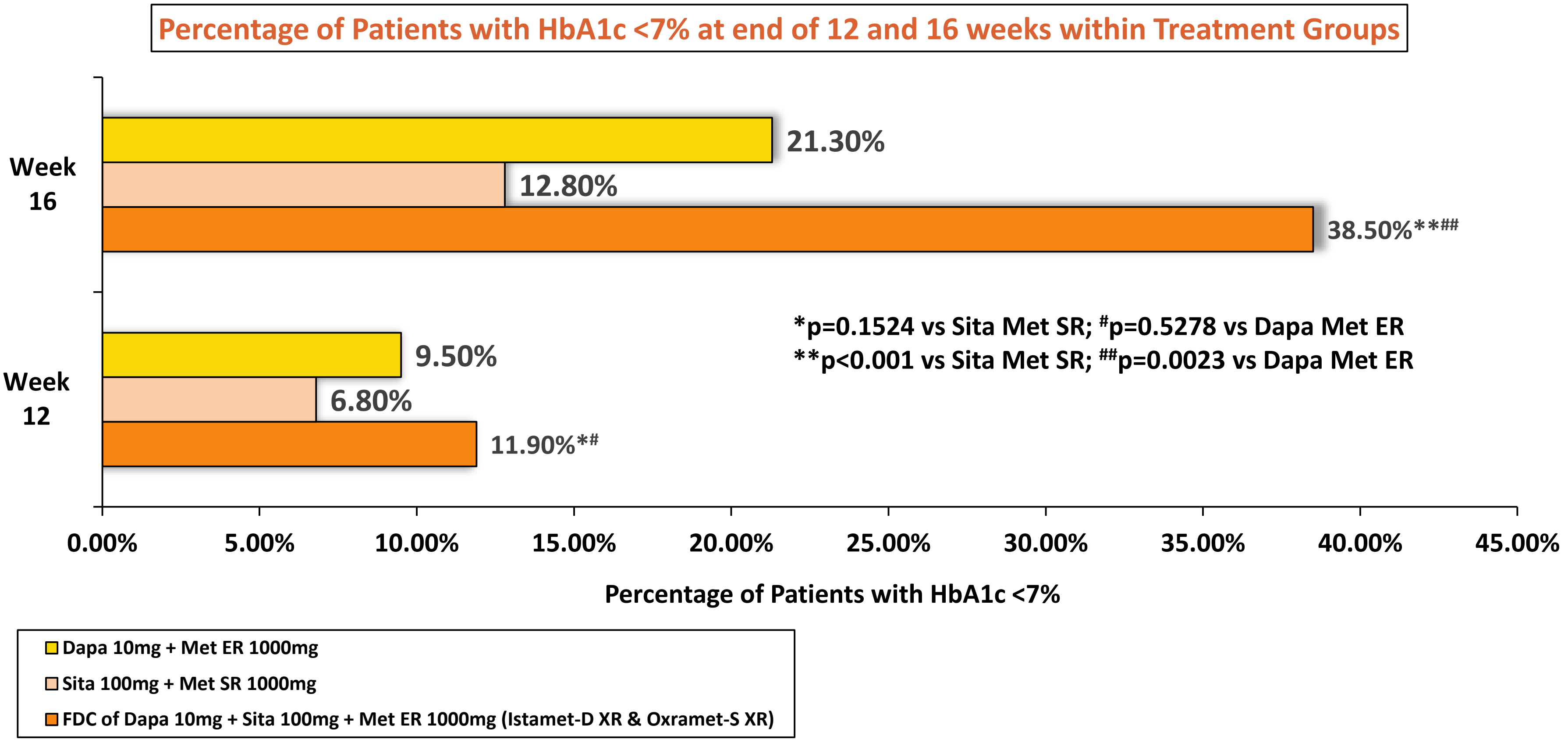
		FDC of Dapa 10 mg +Sita 100 mg +Met ER 1000 mg (N = 137)		Sita 100 mg + Met SR 1000 mg (combipack tablet) (N=139)		FDC of Dapa 10 mg + Met ER 1000 mg (N=139)	
Visits	**Statistic Summary**	**Actual**	**CFB**	**Actual**	**CFB**	**Actual**	**CFB**
Baseline	**n**	137		139		139	
	Mean ± SD	241.8 ± 68.08		237.8 ±57.36		230.8 ± 48.97	
Week 12 (Visit 4)	**n**	135		133		137	
	Mean ± SD	207.3 ± 55.72	**-34.1 ± 52.95**	205.7 ± 57.76	**-30.6 ± 48.57**	202.3 ± 49.43	**-28.2 ± 45.80**
Week 16 (Visit 5)	**n**	130		130		129	
	Mean ± SD	7.30 ± 0.90	**-1.75 ± 0.92***	7.69 ± 0.91	**-1.28 ± 0.69**	7.62 ± 0.92	**-1.32 ± 0.74**
	p value	<0.0001		<0.0001		<0.0001	

Secondary Endpoint: Mean Reduction in FBG (mg/dL) From Baseline to Week 12 and Week 16



		FDC of Dapa 10 mg +Sita 100 mg +Met ER 1000 mg (N = 137)		Sita 100 mg + Met SR 1000 mg (combipack tablet) (N=139)		FDC of Dapa 10 mg + Met ER 1000 mg (N=139)	
Visits	Statistic Summary	Actual	CFB	Actual	CFB	Actual	CFB
Baseline	n	137		139		139	
	Mean ± SD	9.05 ± 0.87		8.99 ± 0.83		8.93 ± 0.86	
Week 12 (Visit 4)	n	135		133		137	
	Mean ± SD	7.90 ± 0.91	**-1.15 ± 0.87**	8.12 ± 0.92	**-0.85 ± 0.70**	8.00 ± 0.91	**-0.94 ± 0.73**
Week 16 (Visit 5)	n	130		130		129	
	Mean ± SD	7.30 ± 0.90	**-1.75 ± 0.92***	7.69 ± 0.91	**-1.28 ± 0.69**	7.62 ± 0.92	**-1.32 ± 0.74**
	p value	<0.0001		<0.0001		<0.0001	

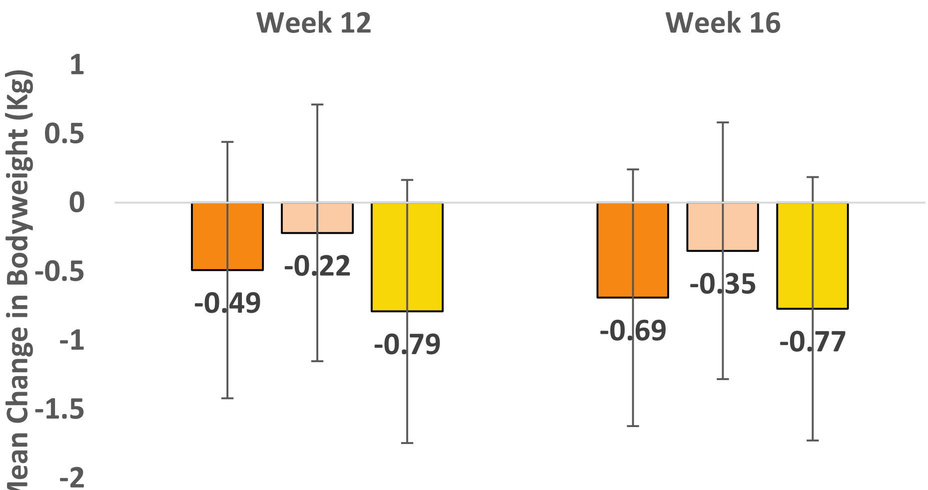
Secondary Endpoint: Proportion of Patients Achieving HbA1c < 7.0% at Week 12 and Week 16



Secondary Endpoint: Mean Change in Bodyweight From Baseline to Week 12 and Week 16

		FDC of Dapa 10 mg +Sita 100 mg +Met ER 1000 mg (N = 137)		Sita 100 mg + Met SR 1000 mg (combipack tablet) (N=139)		FDC of Dapa 10 mg + Met ER 1000 mg (N=139)	
Visits	**Statistic Summary**	**Actual**	**CFB**	**Actual**	**CFB**	**Actual**	**CFB**
Baseline	**n**	137		139		139	
	Mean ± SD	66.41 ± 10.90		67.84 ± 10.99		68.76 ± 11.27	
Week 12 (Visit 4)	**n**	135		133		137	
	Mean ± SD	66.01 ± 10.74	-0.49 ± 1.32	67.72 ± 10.84	-0.22 ± 1.30	67.90 ± 11.21	-0.79 ± 1.97
Week 16 (Visit 5)	**n**	135		135		136	
	Mean ± SD	65.81 ± 10.75	-0.69 ± 1.67	67.59 ± 10.88	-0.35 ± 1.56	67.94 ± 11.32	-0.77 ± 2.15
	p value	<0.0001		0.0105		<0.0001	

Mean reduction in Bodyweight at Week 12 and Week 16



***p= 0.0718 vs Sita Met SR; p=0.1017 vs Dapa Met ER**
****p=0.0245 vs Sita Met SR; p=0.6979 vs Dapa Met ER**

FDC of Dapa 10mg + Sita 100mg + Met ER 1000mg (Istamet-D XR & Oxramet-S XR)

Sita 100mg + Met SR 1000mg

Safety Results (Safety population)

	FDC of Dapagliflozin 10 mg Sitagliptin 100 mg and Extended Release Metformin Hydrochloride 1000 mg tablets (N = 137)		Sitagliptin Phosphate 100 mg and Metformin HCl SR 1000 mg tablets (combipack) (N=139)		FDC of Dapagliflozin 10 mg and Metformin HCl ER tablets 1000 mg (N=139)		Overall (N=415)	
	Events	**n (%)**	**Events**	**n (%)**	**Events**	**n (%)**	**Events**	**n (%)**
Any AEs	24	14 (10.2%)	17	11 (7.9%)	17	13 (9.4%)	58	38 (9.2%)
Any TEAEs	24	14 (10.2%)	17	11 (7.9%)	17	13 (9.4%)	58	38 (9.2%)
Drug-related TEAEs	0	0	0	0	0	0	0	0
Serious TEAEs	0	0	0	0	0	0	0	0

- Overall, 58 TEAEs were reported in 38 (9.2%) patients out of 415 patients.
- All TEAEs were mild in nature.
- Out of 58 TEAEs, 56 events had causality assessment as ‘unlikely’ and remaining two events had ‘conditional/unclassified’ (Abdominal pain upper and Hyperchlorhydria).
- The outcome of all reported TEAEs was resolved at the end of study.
- There were no serious adverse events, deaths, severe or life-threatening TEAEs reported during the study
- <u>One level 1 hypoglycaemia event was reported in one patient in Sitagliptin Phosphate 100 mg and Metformin HCl ER tablets 1000 mg (combipack) arm at randomization visit.</u>
- None of the patient required rescue medications during the study

Treatment Emergent Adverse Events by System Organ Class (SOC) and Preferred Term (PT) - Safety Population

SOC PT	FDC of Dapagliflozin 10 mg Sitagliptin 100 mg and Metformin HCl ER 1000 mg tablets (N = 137)		Sitagliptin Phosphate 100 mg and Metformin HCl ER tablets 1000 mg (combipack) (N=139)		FDC of Dapagliflozin 10 mg and Metformin HCl ER tablets 1000 mg (N=139)		Overall (N=415)	
	n (%)	E	n (%)	E	n (%)	E	n (%)	E
Total TEAEs	***14 (10.2%)***	***24***	***11 (7.9%)***	***17***	***13 (9.4%)***	***17***	***38 (9.2%)***	***58***
Gastrointestinal disorders	**8 (5.8%)**	**14**	**6 (4.3%)**	**7**	**5 (3.6%)**	**6**	**19 (4.6%)**	**27**
Abdominal pain	1 (0.7%)	1	0	0	1 (0.7%)	1	2 (0.5%)	2
Abdominal pain upper	1 (0.7%)	1	0	0	1 (0.7%)	1	2 (0.5%)	2
Diarrhoea	1 (0.7%)	1	3 (2.2%)	3	0	0	4 (1.0%)	4
Dyspepsia	1 (0.7%)	1	0	0	0	0	1 (0.2%)	1
Flatulence	0	0	0	0	1 (0.7%)	1	1 (0.2%)	1
Gastritis	3 (2.2%)	3	1 (0.7%)	1	0		4 (1.0%)	4
Haematochezia	1 (0.7%)	1	0	0	0		1 (0.2%)	1
Hyperchlorhydria	2 (1.5%)	2	1 (0.7%)	1	1 (0.7%)	1	4 (1.0%)	4
Mouth ulceration	0	0	1 (0.7%)	1	0	0	1 (0.2%)	1
Nausea	1 (0.7%)	1	0	0	1 (0.7%)	1	2 (0.5%)	2
Vomiting	3 (2.2%)	3	1 (0.7%)	1	1 (0.7%)	1	5 (1.2%)	5

Treatment Emergent Adverse Events by System Organ Class (SOC) and Preferred Term (PT) - Safety Population_contd.

SOC Patients	FDC of Dapagliflozin 10 mg Sitagliptin 100 mg and Metformin HCl ER 1000 mg tablets (N = 137)		Sitagliptin Phosphate 100 mg and Metformin HCl ER tablets 1000 mg (combipack) (N=139)		FDC of Dapagliflozin 10 mg and Metformin HCl ER tablets 1000 mg (N=139)		Overall (N=415)	
	n (%)	E	n (%)	E	n (%)	E	n (%)	E
General disorders and adm. site conditions	**1 (0.7%)**	**1**	**4 (2.9%)**	**4**	**5 (3.6%)**	**5**	**10 (2.4%)**	**10**
Asthenia	0	0	1 (0.7%)	1	2 (1.4%)	2	3 (0.7%)	3
Chest discomfort	0	0	0	0	1 (0.7%)	1	1 (0.2%)	1
Pain	0	0	0	0	1 (0.7%)	1	1 (0.2%)	1
Pyrexia	1 (0.7%)	1	3 (2.2%)	3	1 (0.7%)	1	5 (1.2%)	5
Infections & infestations	**1 (0.7%)**	**1**	**1 (0.7%)**	**1**	**1 (0.7%)**	**1**	**3 (0.7%)**	**3**
Nasopharyngitis	1 (0.7%)	1	1 (0.7%)	1	1 (0.7%)	1	3 (0.7%)	3
Musculoskeletal & connective tissue disorders	**1 (0.7%)**	**1**	**0**	**0**	**1 (0.7%)**	**1**	**2 (0.5%)**	**2**
Arthralgia	1 (0.7%)	1	0	0	0		1 (0.2%)	1
Back pain	0	0	0	0	1 (0.7%)	1	1 (0.2%)	1
Nervous system disorders	**2 (1.5%)**	**2**	**1 (0.7%)**	**1**	**3 (2.2%)**	**3**	**6 (1.4%)**	**6**
Burning sensation	1 (0.7%)	1	0	0	0	0	1 (0.2%)	1
Dizziness	0	0	1 (0.7%)	1	0	0	1 (0.2%)	1
Headache	1 (0.7%)	1	0	0	3 (2.2%)	3	4 (1.0%)	4

Treatment Emergent Adverse Events by System Organ Class (SOC) and Preferred Term (PT) - Safety Population_contd.

SOC Patients	FDC of Dapagliflozin 10 mg Sitagliptin 100 mg and Metformin HCl ER 1000 mg tablets (N = 137)		Sitagliptin Phosphate 100 mg and Metformin HCl ER tablets 1000 mg (combipack) (N=139)		FDC of Dapagliflozin 10 mg and Metformin HCl ER tablets 1000 mg (N=139)		Overall (N=415)	
	n (%)	E	n (%)	E	n (%)	E	n (%)	E
Psychiatric disorders	**0**	**0**	**1 (0.7%)**	**1**	**0**	**0**	**1 (0.2%)**	**1**
Anxiety	0	0	1 (0.7%)	1	0	0	1 (0.2%)	1
Renal and urinary disorders	**1 (0.7%)**	**1**	**0**	**0**	**0**	**0**	**1 (0.2%)**	**1**
Urine abnormality	1 (0.7%)	1	0	0	0	0	1 (0.2%)	1
Reproductive system and breast disorders	**0**	**0**	**1 (0.7%)**	**1**	**0**	**0**	**1 (0.2%)**	**1**
Vulvovaginal candidiasis	0	0	1 (0.7%)	1	0	0	1 (0.2%)	1
Respiratory, thoracic and mediastinal disorders	**1 (0.7%)**	**1**	**0**	**0**	**1 (0.7%)**	**1**	**2 (0.5%)**	**2**
Cough	1 (0.7%)	1	0	0	0	0	1 (0.2%)	1
Sneezing	0	0	0	0	1 (0.7%)	1	1 (0.2%)	1
Skin and subcutaneous tissue disorders	**1 (0.7%)**	**2**	**1 (0.7%)**	**2**	**0**	**0**	**2 (0.5%)**	**4**
Erythema	1 (0.7%)	1	0	0	0	0	1 (0.2%)	1
Pruritus	1 (0.7%)	1	1 (0.7%)	1	0	0	2 (0.5%)	2
Rash	0	0	1 (0.7%)	1	0	0	1 (0.2%)	1
Eye disorders	**1 (0.7%)**	**1**	**0**	**0**	**0**	**0**	**1 (0.2%)**	**1**
Ocular hyperaemia	1 (0.7%)	1	0	0	0	0	1 (0.2%)	1

Study Conclusion

- FDC of Dapagliflozin 10 mg, Sitagliptin 100 mg and Metformin HCl ER 1000 mg was superior in comparison to both two drug combinations (Sitagliptin phosphate 100 mg and Metformin HCl SR 1000 mg [combipack] tablets; and FDC of Dapagliflozin 10 mg and Metformin HCl ER 1000 mg tablets) in terms of HbA1c reduction at Week 12 and Week 16
- The study drugs were safe and well tolerated

Thank you